# Executive Report: Waze User Churn Inspection & Preparation

Project: User Churn Prediction Model | Client: Waze Data Strategy Team
PACE Phase: Analyze (Data Inspection and Structuring)
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## OVERVIEW

The primary goal of this project is to identify the factors that influence user churn at Waze. By developing a predictive model, we aim to understand why users stop using the app and provide actionable insights to improve retention. At this stage, we have completed a deep dive into the dataset to ensure data integrity and identify initial behavioral patterns.

## Business Questions

IIn this initial stage, we address specific technical and operational questions to guide the model's development:

- What is the distribution of churned vs. retained users?
- Are there significant differences in driving intensity (km per day) between these groups?
- How does device type (Android vs. iPhone) impact the likelihood of churn?
- What is the profile of the most "active" users who are leaving the platform?

## Project Overview

We are transforming raw user-generated data into a structured format for machine learning. Our current workflow includes:

- Data Structuring: Built a dataframe with 14,999 rows and identified key variables like sessions, driving_days, and driven_km_drives.
- Feature Engineering: Created new indicators such as km_per_driving_day and drives_per_driving_day to better capture user intensity.
- Anomaly Detection: Identified extreme outliers in driving distances (e.g., users driving >20,000 km/month) and used median statistics to ensure robust conclusions.
- Missing Value Analysis: Confirmed that 700 missing labels were missing at random (MCAR) and handled them to prevent model bias.

## PACE Methodology

We have adopted the PACE framework to ensure a disciplined data science approach:

### Plan:

- Defined project goals: Building a model to predict and reduce user churn.
- Identified key stakeholders and data dictionary requirements.

### Analyze:

- Focused on data inspection, finding that churned users drive 2.4x more distance per day than retained users (~697 km vs ~289 km).
- Determined that device type is not a factor, as churn rates are identical for iPhone and Android.
- A critical finding during this phase is that churned users are NOT inactive; instead, they exhibit an 'extremely high-intensity' profile. While they use the app on fewer days per month (median of 6 days vs. 14 for retained users), they drive significantly more—averaging nearly 700 km and 10 separate drives per active day. This data suggests that the churned group likely consists of professional or long-haul drivers whose specific needs (e.g., specialized routing, long-duration stability) may not be currently met by the standard commuter-focused application experience.

### Construct (Upcoming):

- Conducting full Exploratory Data Analysis (EDA) and selecting features for the classification model.
- Validating the "Professional Driver" hypothesis through visualizations.

### Execute (Upcoming):

- Deployment of the churn prediction tool and reporting final strategy recommendations to management.

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## IMPACT

By identifying behavioral anomalies and organizing the data early, we ensure a high-quality "Tidy Data" foundation. This reduces future errors in the churn prediction modeling phase and provides a clear roadmap for a Deep Dive Exploratory Data Analysis (EDA), ultimately leading to more precise retention strategies for the platform.

## Key Insights

AAfter inspecting the variables in the dataset, we identified that driving intensity and frequency are the most critical factors for the initial model. Preliminary inspection revealed significant outliers and a unique high-usage churn profile.

### Anomalous Data Reference

- "Super-Driver" Profile: We identified "unusual values" where churned users drive disproportionately high distances (~697 km/day) compared to retained users. This indicates a high-intensity segment that uses the app heavily but only for a few days.
- Skewed Distribution: The dataset shows that driving distances and session durations are heavily right-skewed, requiring the use of median statistics to avoid model bias.
- Behavioral Correlation: There is a strong correlation between high daily trip volume and churn. Churned users have 2.5x more drives per driving day (median) than retained users.
- Platform Neutrality: Normalized device metrics show no correlation between mobile OS (iPhone vs. Android) and churn, shifting the focus from technical issues to user experience and utility.

| User Group | Driving Days (Median) | Km per Driving Day (Median) | Drives per Driving Day (Median) |
|---|---|---|---|
| **Retained** | 14 days | 289.5 km | 4.0 drives |
| **Churned** | 6 days | 697.5 km | 10.0 drives |

"...suggesting that extreme usage intensity over short periods is a primary indicator for prioritized churn intervention."

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## Project Governance & Roles

To ensure project success, we have defined specific roles and responsibilities aligned with the PACE framework, ensuring clear communication between technical teams and institutional stakeholders.

| Role | Assignee | Responsibility |
|---|---|---|
| Data Analysis Director | May Santner | Strategic oversight, final approval of insights, and project direction. |
| Senior Data Analyst | Chidi Ga | Technical mentorship, review of EDA, and quality assurance of data processes. |
| Senior Project Manager | Sylvester Esperanza | Project coordination, timeline management, and stakeholder alignment. |
| Data Analyst | Marcelo Dominguez | Data cleaning, statistical analysis, and initial model construction. |
| Data Scientist | Harriet Hadzic | Advanced modeling support and machine learning strategy. |

## Next Steps

### Deep-Dive Exploratory Data Analysis **(EDA)**:

- Conduct a deep dive into the validated dataset to identify core correlations between driving intensity, sessions, and churn status.
- Create visual distributions (Boxplots/Histograms) to compare the 700 km/day segment against typical commuters.

### Advanced Feature Engineering:

- Prioritize variables with a direct impact on classification. We will evaluate "Data Boosters" like driving efficiency ratios and session frequency to enrich the model's predictive power before training.

### Model Selection & Evaluation:

- Evaluate Random Forest and XGBoost as primary classification candidates. We will compare performance metrics (Precision/Recall) to select the most robust algorithm for automated churn detection.

### Documentation & Visual Strategy:

- Update the project repository on GitHub and design specialized dashboards to communicate insights to both technical and non-technical stakeholders (May Santner & Chidi Ga).

*""Our analysis reveals a critical link between usage intensity and churn. While device types are evenly split, churned users generate significantly higher daily mileage. Most importantly, the median daily distance for churned users is 2.4x higher than for retained users. This suggests that high-intensity driving patterns are the strongest predictors for identifying users at risk of leaving the platform."*